AIRLINE BUSINESS INTELLIGENCE & REVENUE ANALYTICS SYSTEM

Executive Summary

The Airline Business Intelligence & Revenue Analytics System project aims to analyze airline pricing behavior, route profitability, booking patterns, and revenue performance using Python, SQL, and Power BI. The objective of this project is to derive actionable business insights that can help airlines optimize pricing strategies, improve route profitability, and enhance revenue generation.
The analysis was performed on a dataset containing more than 300,000 flight records across multiple airlines, routes, travel classes, and booking periods.

Business Problem Statement

The airline industry faces challenges in:

- Optimizing ticket pricing strategies
- Understanding customer booking behavior
- Identifying profitable routes
- Maximizing airline revenue
- Improving business decision-making

This project addresses these challenges by performing comprehensive business intelligence and revenue analytics.

Project Objectives

The primary objectives of this project are:

- Analyze airline pricing trends
- Identify the highest revenue-generating airlines
- Determine the most profitable routes
- Analyze customer booking behavior
- Evaluate business versus economy class performance
- Measure airline market share
- Generate actionable business recommendations

Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SQL (MySQL)
- Power BI
- GitHub

Dataset Overview

- Total Records: 300,153
- Total Airlines: Multiple major airlines
- Total Routes: 100+ routes
- Data Fields: Airline, Flight, Source City, Destination City, Stops, Duration, Days Left, Travel Class, Ticket Price

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Key Business Questions

The project focused on answering the following business questions:

1. Which airlines generate the highest revenue?
2. Which routes are the most profitable?
3. How does booking timing affect ticket prices?
4. Which travel class contributes the most revenue?
5. Which cities generate the highest revenue?
6. Which airlines follow premium pricing strategies?
7. What is the airline market share distribution?
8. How do flight stops impact ticket pricing?

Major Findings

Airline Revenue Analysis

- Premium airlines generate significantly higher revenue.
- Market share is concentrated among a small number of major airlines.

Route Profitability Analysis

- Certain routes contribute disproportionately to total revenue.
- High-demand metropolitan routes generate maximum profits.

Booking Behavior Analysis

- Last-minute bookings are generally more expensive.
- Advance booking provides lower ticket prices.

Travel Class Analysis

- Business class contributes a significant portion of revenue despite lower booking volume.
- Economy class dominates total passenger traffic.

Pricing Strategy Analysis

- Airlines adopt different pricing strategies based on competition, demand, and route popularity.

SQL Analytics Performed

Advanced SQL techniques used:

- Common Table Expressions (CTEs)
- Window Functions
- DENSE_RANK()
- Revenue Contribution Analysis
- Market Share Analysis
- Pricing Strategy Analysis
- Booking Behavior Analysis
- Route Profitability Analysis

Power BI Dashboard

The dashboard consists of:

Executive Dashboard

- Total Revenue
- Total Flights
- Average Ticket Price
- Total Routes
- Total Airlines

Airline Analysis Dashboard

- Revenue by Airline
- Market Share Analysis

- Average Ticket Price
- Airline Performance Summary

Business Recommendations

Based on the analysis, the following recommendations are proposed:

1. Focus operations on highly profitable routes.
2. Expand premium business-class offerings.
3. Implement dynamic pricing strategies.
4. Encourage advance booking through promotional offers.
5. Optimize operations on low-performing routes.
6. Improve revenue management through demand forecasting.
7. Increase market share in high-demand cities.

Conclusion

The Airline Business Intelligence & Revenue Analytics System successfully identified key revenue drivers, customer behavior patterns, and route profitability trends. The project demonstrates practical application of data analytics, SQL, business intelligence, and dashboarding techniques that support strategic business decision-making in the airline industry.